

Tomás Dias | Software Developer

MSc Candidate in Computer Science and Engineering

Almada, Portugal

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Summary

MSc candidate in Computer Science and Engineering specializing in **Artificial Intelligence**, with a minor in **Systems and Software Security**. Curious and adaptable, with a strong interest in solving complex problems and learning new technologies. Focused on developing reliable, well-designed software for technically challenging projects.

Skills

Languages: Java, Python, C, SQL, JavaScript

AI & Data: PyTorch, Scikit-Learn, NumPy, Pandas, Clingo (ASP), OWL 2

Backend: Quarkus, Apache Kafka, REST APIs, Node.js

Databases: Neo4j (Cypher), Oracle, H2, SPARQL

Tools & Practices: Docker, Git, Linux, Jira, Agile (Scrum)

Spoken Languages: Portuguese (Native), English (Fluent)

Work Experience

Layer8

Software Engineering Intern – R&D

Lisbon, Portugal

Feb 2024 – Jul 2024

- Engineered an automated ingestion pipeline in Java (Quarkus), dynamically parsing STIX and YAML feeds to model MITRE ATT&CK techniques and Atomic Red Team tests.
- Built an event-driven architecture on Apache Kafka, with REST endpoints for real-time querying.
- Architected a Neo4j graph using Cypher to map CVE vulnerabilities and attack pattern dependencies.

Relevant Projects

Recovering Genomes from Microbial Communities with Domain Constraints

MSc Dissertation – Python, Clingo, ASP

2025–2026

- Developed an Answer Set Programming (ASP) framework in Clingo to reconstruct microbial genomes from sequence fragments under biological constraints.
- Redesigned prior ASP encodings by replacing exhaustive pairwise matrices with a sparse nearest-neighbor affinity graph, eliminating quadratic grounding bottlenecks and increasing performance.
- Architected a modular refinement pipeline using targeted separation, merge, and recruitment operators to repair chimeric and over-split clusters often generated by standard heuristic binning tools.

Concurrent B-Tree Synchronization

Java, Concurrency

2025

- Engineered a concurrent B-Tree in Java, implementing proactive top-down splitting to ensure a deadlock-free architecture by construction.
- Developed fine-grained lock coupling and an optimistic search strategy with volatile version validation for fast lock-free reads.
- Benchmarked coarse-grained, lock-coupled and optimistic variants up to 16 threads, validating correctness with stress tests against a ConcurrentHashMap oracle and a structural-invariant checker.

Semi-Supervised Survival Prediction (Multiple Myeloma)

Python, Scikit-Learn

2024

- Formulated survival time prediction as a semi-supervised ML regression task using clinical and genetic data.
- Handled missing clinical features with MICE imputation and unobserved outcomes from right-censored data via a censored MSE loss.
- Benchmarked regularized linear models against gradient-boosted tree ensembles (HistGradientBoosting, CatBoost).

Education

NOVA School of Science and Technology

MSc in Computer Science and Engineering

Almada, Portugal

2024–2026 (Expected)

Major: Artificial Intelligence | Minor: Systems and Software Security

Dissertation: *Recovering Genomes from Microbial Communities with Domain Constraints*

Relevant Coursework: Machine Learning, Deep Learning, Computer Systems Security, Software Security, Concurrent Systems, Knowledge Representation, Autonomous Decision Making, Bioinformatics, Human-Computer Interaction.

NOVA School of Science and Technology

BSc in Computer Science and Engineering

Almada, Portugal

2021–2024

Relevant Coursework: Data Structures & Algorithms, Distributed Systems, Operating Systems, Computer Networks, Databases, Object-Oriented Programming (OOP), Software Engineering, Probability & Statistics.